E-COMMERCE PLATFORM FOR PERSONALIZED RETAIL EXPERIENCES

Project Group: 25-26J-526

Team Members & Components

- IT22588814 (Pramuditha W. G. S.) - Sinhala Voice Assistant
- IT22367044 (Nayanapriya H. T. E.) - Personalized Promotion Engine
- IT22326072 (Nimsara S. P. N.) - Customer Behavioral Segmentation
- IT22306968 (Dinuwan W. T.) - BI Dashboard with XAI

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Research Gap

- Most e-commerce voice systems mainly support **English**, with limited support for **Sinhala language and regional accents**.
- Existing promotion systems use **generic offers** and do not consider **customer behavior or context**.
- Many **Business Intelligence tools are complex and expensive**, making them less accessible for **SMEs**.
- AI-driven recommendations often lack **explainability**, reducing **trust and transparency** for business users.
- Current **customer segmentation and loyalty systems are static** and do not adapt to **changing customer behavior**.

Proposed Solution

- AI-powered e-commerce platform for personalized shopping.
- Sinhala Voice Assistant for natural voice-based shopping.
- Personalized Promotion Engine based on customer behavior.
- Dynamic customer segmentation to improve targeting.
- BI Dashboard with Explainable AI for real-time insights.

Who is affected & Who Benefits

Customers

- Sinhala users
- Elderly and accessibility
- focused users

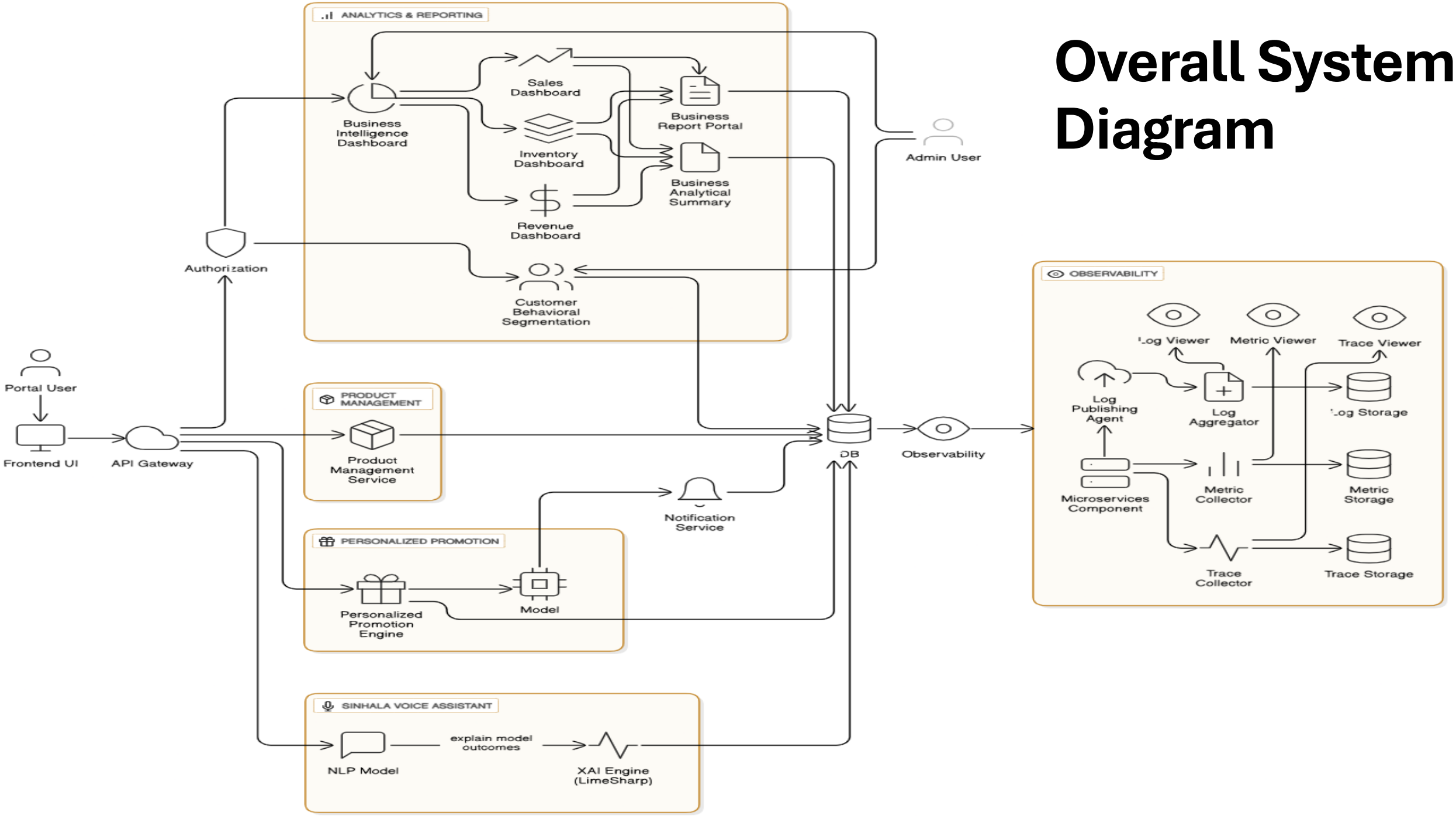
Small & Medium Enterprises (SMEs)

- Local retailers

Other Benefits

- Reduced digital divide
- Increased participation of local businesses
- Realtime, explainable BI insights
- Smarter promotions with measurable uplift
- Adaptive loyalty systems that grow customer lifetime value

Overall System Diagram



E-Commerce Platform for Personalized Retail Experiences

Commercialization

- SaaS platform with mobile app support
- Tiered pricing packages (Basic, Pro, Enterprise)
- Enterprise licensing with customization and support

Customer Persona

- Retail shop owners (small to large)
- E-commerce platforms seeking accessibility features
- Businesses targeting rural Sri Lankan markets

Target Market & Customers

- Sri Lankan retail market: $15B+
- Target segment: 5,000+ medium to large retail stores

What is Unique in Your Solution

- First-mover in Sinhala e-commerce voice AI integration
- Bilingual XAI (Sinhala + English) for retail forecasting
- 40% improvement in rural user engagement
- Scalable for other low-resource languages

How will the cost be recovered?

- Enterprise licensing and customization fees
- Covers labor, infrastructure, AI APIs, and security

Thank You!